Applied Microprocessor Lab

2023 Fall

Final Term Project
Example: OpenCL with JNI

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Final Term Project

- **Make an android application processing an image taken from camera**
  1. **Use OpenCL kernel on JNI**
  2. **Provide several image processing methods**

     e.g., Blur, Grayscale, rotation, color inversion, etc.
  3. **Use I/O devices**

     e.g., 7 segment display, LEDs, etc.
  4. **Add your idea**

Important Dates

- **Final Exam**
  - Date: 12/06 (Wed) 1:00 pm
  - Location: TBD
- **Final Presentation (Term project)**
  - Date: 12/20 (Wed) 1:00 pm
  - Location: R213
  - Individual Project
    - Presentation: 6 min per person
    - Introduction + Demo
  - Submit Documents
    - Template will be provided

OpenCL on Android

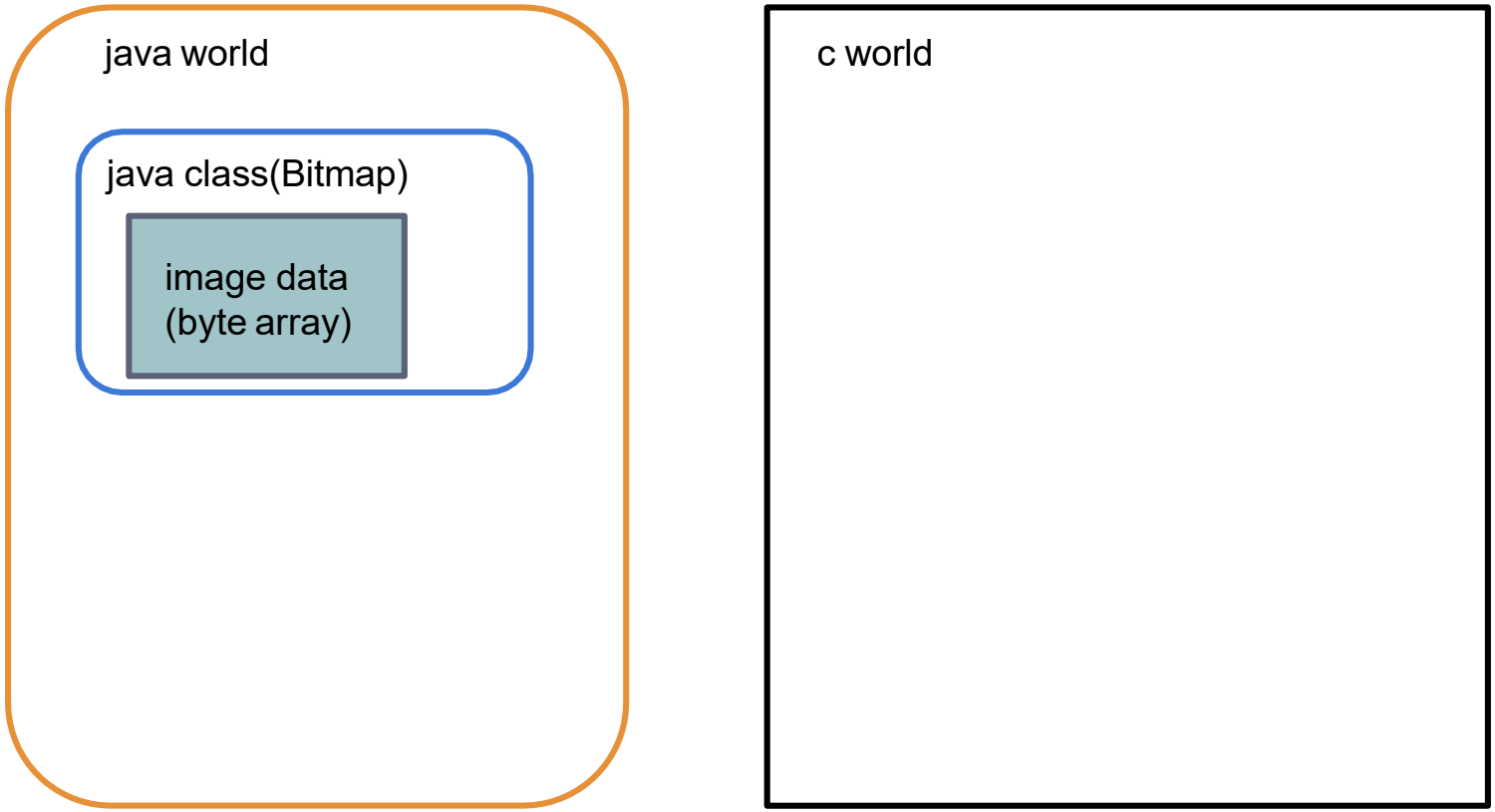
- **Gaussain Blur app with JNI**

OpenCL on Android

- **Materials**
  1. Android Studio
  2. Bitmap Image file

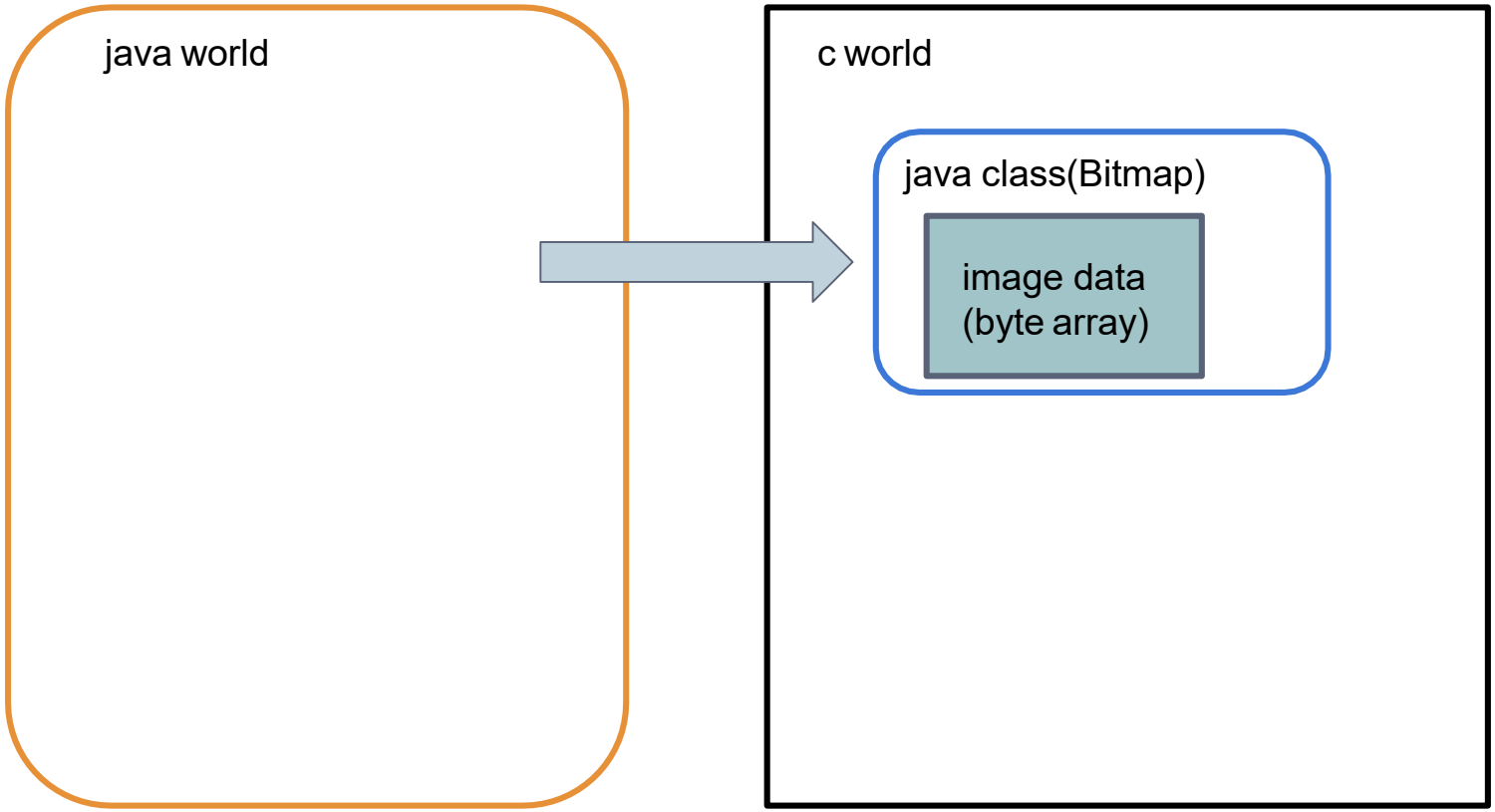
OpenCL on Android

- **Read image on Java code**



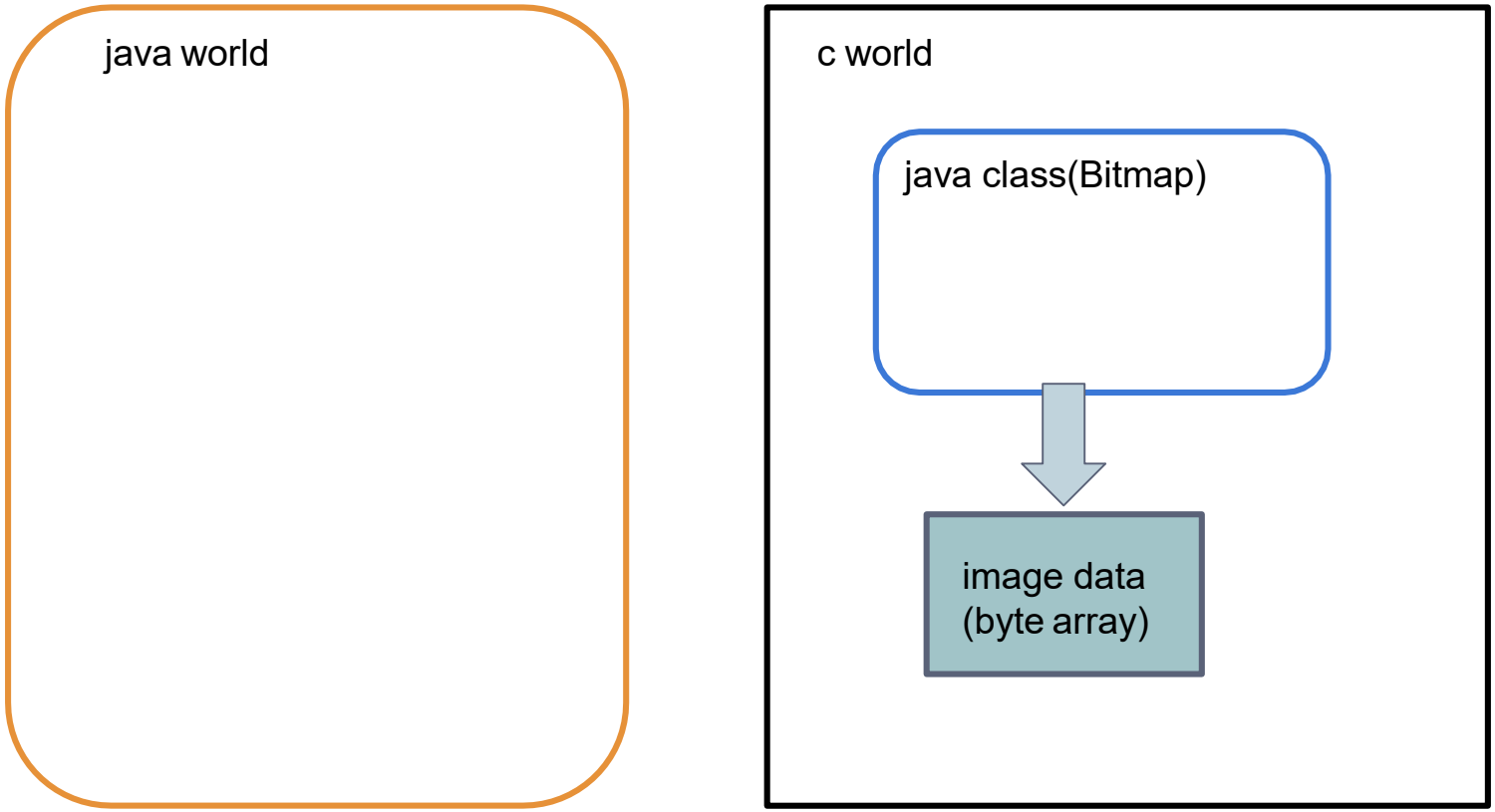
OpenCL on Android

- **Send image object to C world (JNI)**



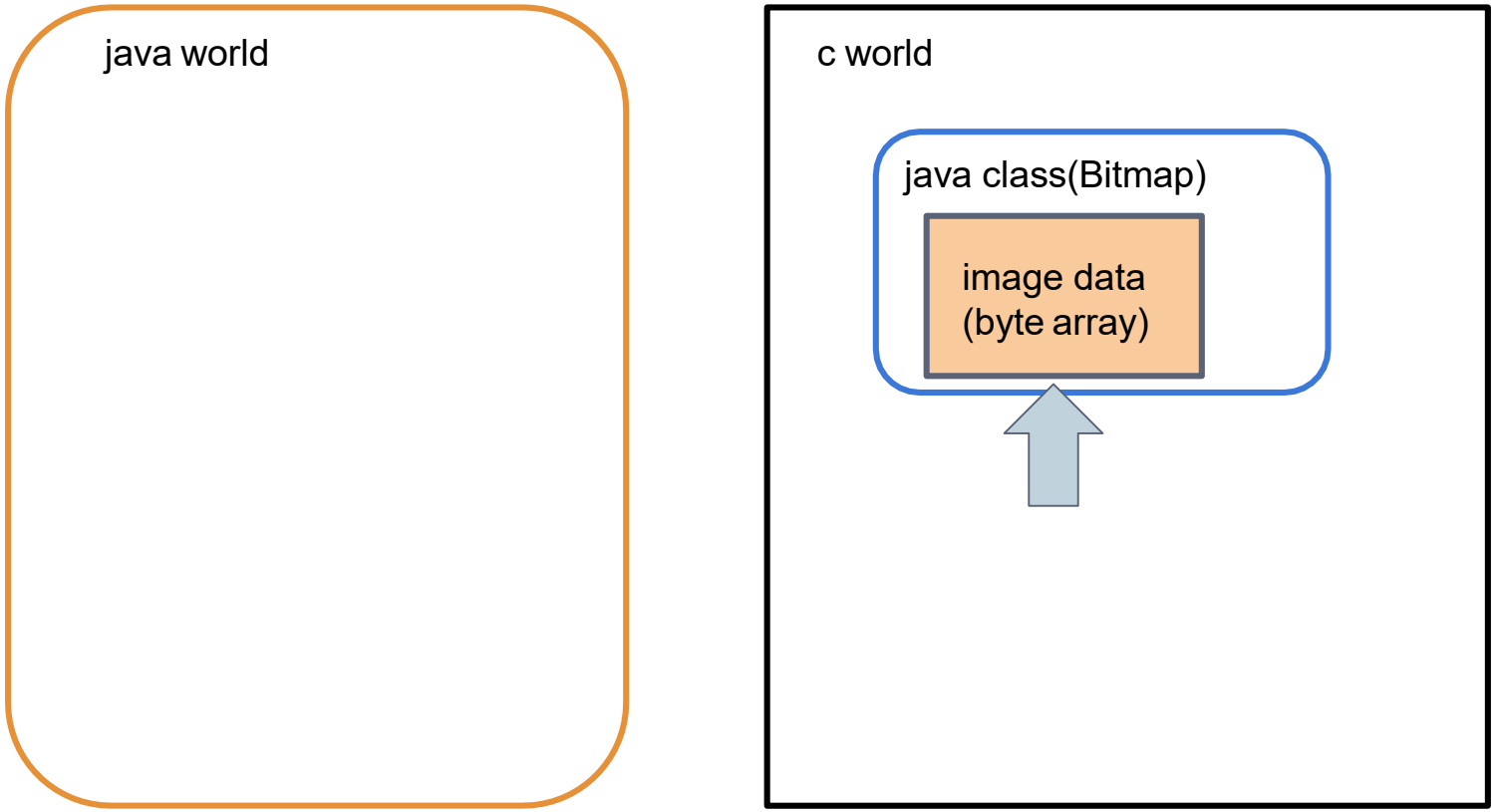
OpenCL on Android

- **Copy image**



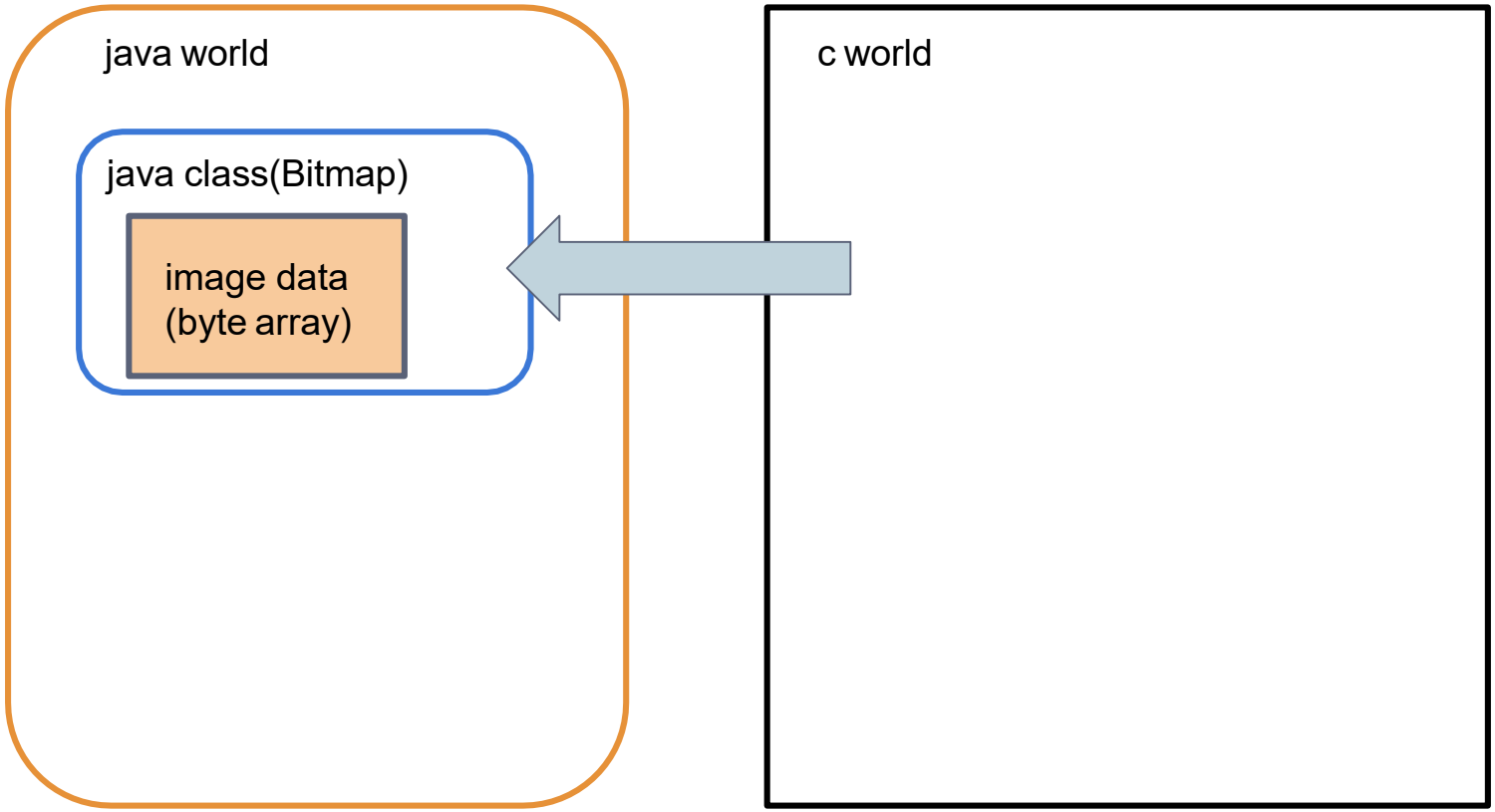
OpenCL on Android

- **Process the image using OpenCL and copy it back**



OpenCL on Android

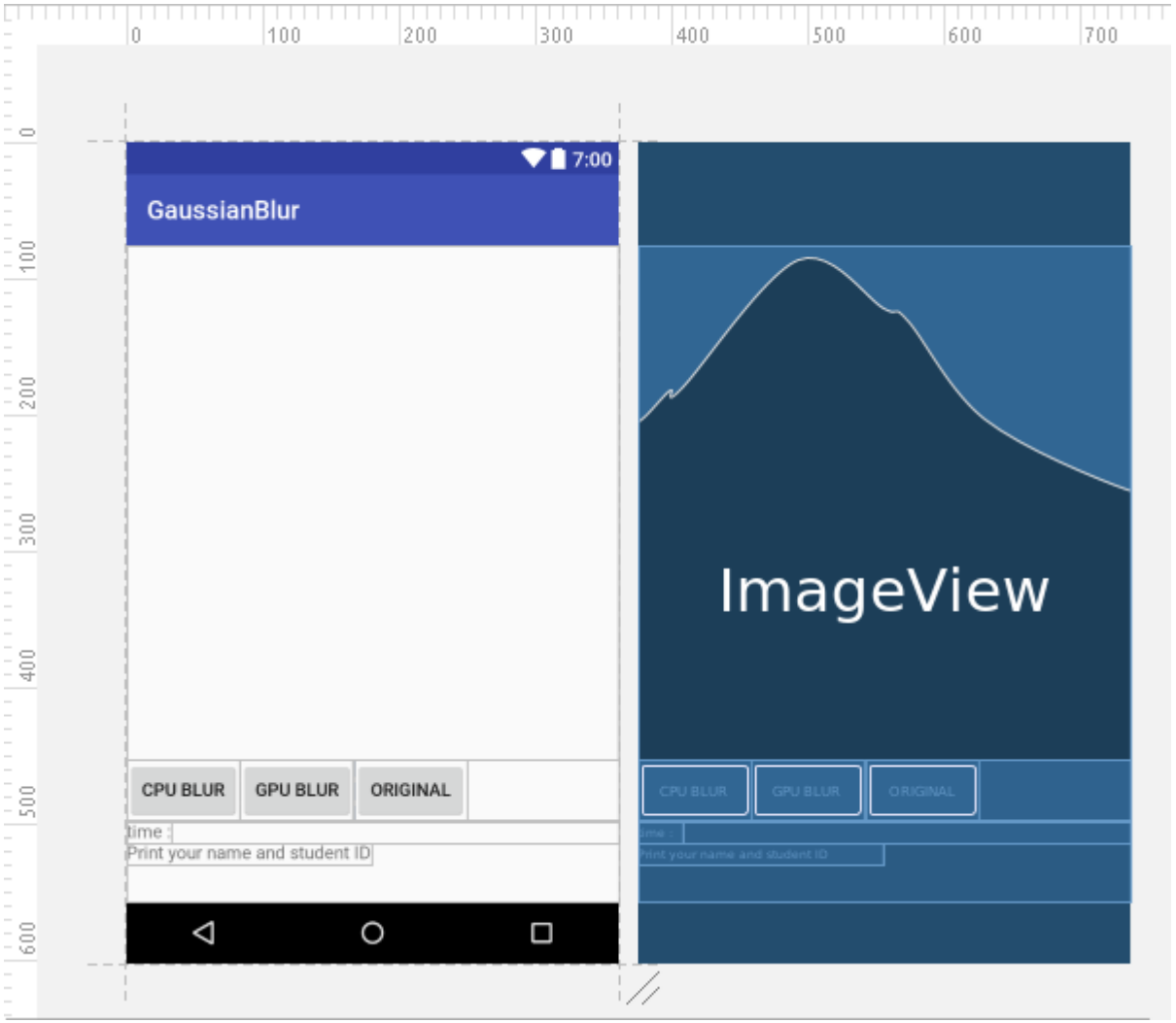
- **Send back the image to Java world**



OpenCL on Android

- **Files**
  1. lena.bmp
  2. MainActivity.java
  3. activity_main.xml
  4. OpenCLDriver.c
  5. CMakeLists.txt

OpenCL
activity_main.xml



OpenCL MainActivity.java (1/7)

- **Load native library and native functions**

```
static {
    System.loadLibrary("OpenCLDriver");
}
//blur CPU
public native Bitmap GaussianBlurBitmap(Bitmap bitmap);
//blur GPU
public native Bitmap GaussianBlurGPU(Bitmap bitmap);
```

OpenCL MainActivity.java (2/7)

- **Declare variables of MainActivity class**

```
22
23      ImageView imgV;
24      Bitmap buf_bitmap;
25
```

OpenCL MainActivity.java (3/7)

- **Override onCreate()**

```
26      @Override
27      protected void onCreate(Bundle savedInstanceState) {
28          super.onCreate(savedInstanceState);
29          setContentView(R.layout.activity_main);
30
31          final TextView tv =(TextView) findViewById(R.id.txt);
32          imgV =(ImageView) findViewById(R.id.imageView);
33          Button btn = (Button) findViewById(R.id.button);
34          Button btn2 = (Button) findViewById(R.id.button2);
35          Button btn3 = (Button) findViewById(R.id.button3);
36
37          BitmapFactory.Options options = new BitmapFactory.Options();
38          options.inPreferredConfig = Bitmap.Config.ARGB_8888;
39
40
41          buf_bitmap=BitmapFactory.decodeFile("/data/local/tmp/lena.bmp", options);
42
43          tv.setText("Image Processing App"  );
44
45          imgV.setImageBitmap(buf_bitmap);
46
```

read image from the path

OpenCL MainActivity.java (4/7)

- **set onClickListener to buttons**

```
btn.setOnClickListener(new View.OnClickListener(){
        //setImageBitmap with BLURRED IMG processed with CPU
});

btn2.setOnClickListener(new View.OnClickListener(){
        //setImageBitmap with BLURRED IMG processed with GPU
});

btn3.setOnClickListener(new View.OnClickListener(){
        //setImageBitmap with the ORIGINGAL IMAGE
});
```

OpenCL MainActivity.java (5/7)

- **set onClickListener to buttons**

```
btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        //When this button is clicked,
        //Process the Bitmap class using native function
                                                GaussianBlurBitmap()
        //then set the result Bitmap image to the ImageView
        //and print the execution time on TextView
    }
});
```

OpenCL MainActivity.java (6/7)

- **set onClickListener to buttons**

```
btn2.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        //When this button is clicked,
        //Process the Bitmap class using native function
                                                GaussianBlurGPU()
        //then set the result Bitmap image to the ImageView
        //and print the execution time on TextView
    }
});
```

OpenCL MainActivity.java (7/7)

- **set onClickListener to buttons**

```
btn3.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        BitmapFactory.Options options = new BitmapFactory.Options();
        options.inPreferredConfig = Bitmap.Config.ARGB_8888;
        buf_bitmap =
                BitmapFactory.decodeFile("/data/local/tmp/lena.bmp",
                                                        options);
        imgV.setImageBitmap(buf_bitmap);
        tv.setText("Orignal image");
    }
});
```

OpenCL MainActivity.java

- **How to measure execution time of the code in android (Java)**

```
float start = (float)System.nanoTime()/1000000;
 //code to measure
 //……
 //…...
float end = (float)System.nanoTime()/1000000;
float timesub = end-start;

String ts = Float.toString(timesub);
textView.setText("Execution Time: " + ts)
```

OpenCL
OpenCLDriver.c (1/5)

- **recommended imports**

```
4   #include <jni.h>
5   #include <stdio.h>
6   #include <stdlib.h>
7   #include <sys/time.h>
8   #include <android/log.h>
9   #include <android/bitmap.h>
10
11  #include <CL/opencl.h>
12
```

OpenCL
OpenCLDriver.c (2/5)

- **You should implement native functions**

```
#define  LOG_TAG    "DEBUG"
#define  LOGD(...)  __android_log_print(ANDROID_LOG_DEBUG,LOG_TAG,__VA_ARGS__)
#define  LOGE(...)  __android_log_print(ANDROID_LOG_ERROR,LOG_TAG,__VA_ARGS__)

JNIEXPORT jobject JNICALL
Java_com_example_psdlab_gaussianblur_MainActivity_GaussianBlurBitmap
       (JNIEnv *env, jclass  class , jobject bitmap)
{
   //this native function is GaussianBlur bitmap image using CPU
}


JNIEXPORT jobject JNICALL
Java_com_example_psdlab_gaussianblur_MainActivity_GaussianBlurGPU
       (JNIEnv *env, jclass  class, jobject bitmap )
{
    //this native function is GaussianBlur bitmap image using OpenCL kernel
}
```

OpenCL
OpenCLDriver.c (3/5)

- **native function handling Bitmap java class**

```
//getting bitmap info:
LOGD("reading bitmap info...");
AndroidBitmapInfo info;
int ret;
if ((ret = AndroidBitmap_getInfo(env, bitmap, &info)) < 0)
{
   LOGE("AndroidBitmap_getInfo() failed ! error=%d", ret);
   return NULL;
}
LOGD("width:%d height:%d stride:%d", info.width, info.height, info.stride);
if (info.format != ANDROID_BITMAP_FORMAT_RGBA_8888)
{
   LOGE("Bitmap format is not RGBA_8888!");
   return NULL;
}
```

OpenCL
OpenCLDriver.c (4/5)

- **native function handling Bitmap java class**

```
//read pixels of bitmap into native memory :
LOGD("reading bitmap pixels...");
void* bitmapPixels;
if ((ret = AndroidBitmap_lockPixels(env, bitmap, &bitmapPixels)) < 0)
{
   LOGE("AndroidBitmap_lockPixels() failed ! error=%d", ret);
   return NULL;
}

uint32_t* src = (uint32_t*) bitmapPixels;
uint32_t* tempPixels = (uint32_t*)malloc(info.height * info.width*4);

int pixelsCount = info.height * info.width;
memcpy(tempPixels, src, sizeof(uint32_t) * pixelsCount);
```

OpenCL
OpenCLDriver.c (5/5)

- **native function handling Bitmap java class**

```
// write image processing code here!!
// input is tempPixels
// output is bitmapPixels <- access via ‘src’ pointer
```

```
AndroidBitmap_unlockPixels(env, bitmap);
//
// free the native memory used to store the pixels
//
free(tempPixels);
return bitmap;
```

OpenCL CMakeLists.txt (1/2)

- **set the library in CMakeLists**

```
add_library(GLES_mali  STATIC  IMPORTED )


add_library( # Sets the name of the library.
             OpenCLDriver

             # Sets the library as a shared library.
             SHARED

             # Provides a relative path to your source file(s).
             OpenCLDriver.c  )
```

```
find_library( # Sets the name of the path variable.
              jnigraphics-lib

              # Specifies the name of the NDK library that
              # you want CMake to locate.
              jnigraphics )
```

OpenCL CMakeLists.txt (2/2)

- **Link libraries**

```
                        ${log-lib} )
target_link_libraries( # Specifies the target library.
                        OpenCLDriver

                        # Links the target library to the log library
                        # included in the NDK.
                        /home/psdlab/OpenCL_files/lib/libGLES_mali.so ${log-lib} ${jnigraphics-lib} )
```

the path of **libGLES_mali.so**

```
target_include_directories(OpenCLDriver PRIVATE src/main/cpp)
```

OpenCL on Android

*** You can use the Gaussian Blur functions and OpenCL kernel code that you have implemented in the previous Lab.**

But you must modify the code because Bitmap image in Android uses **RGBA** format instead of RGB format.

img[0]: R
img[1]: G
img[2]: B
img[3]: **A** == Alpha channel (Opacity)

255: **visible**, 0: **invisible**, between 0 and 255: **translucence**